ITE Arbitrator: A Reference Architecture Framework for Sustainable IT Ecosystems

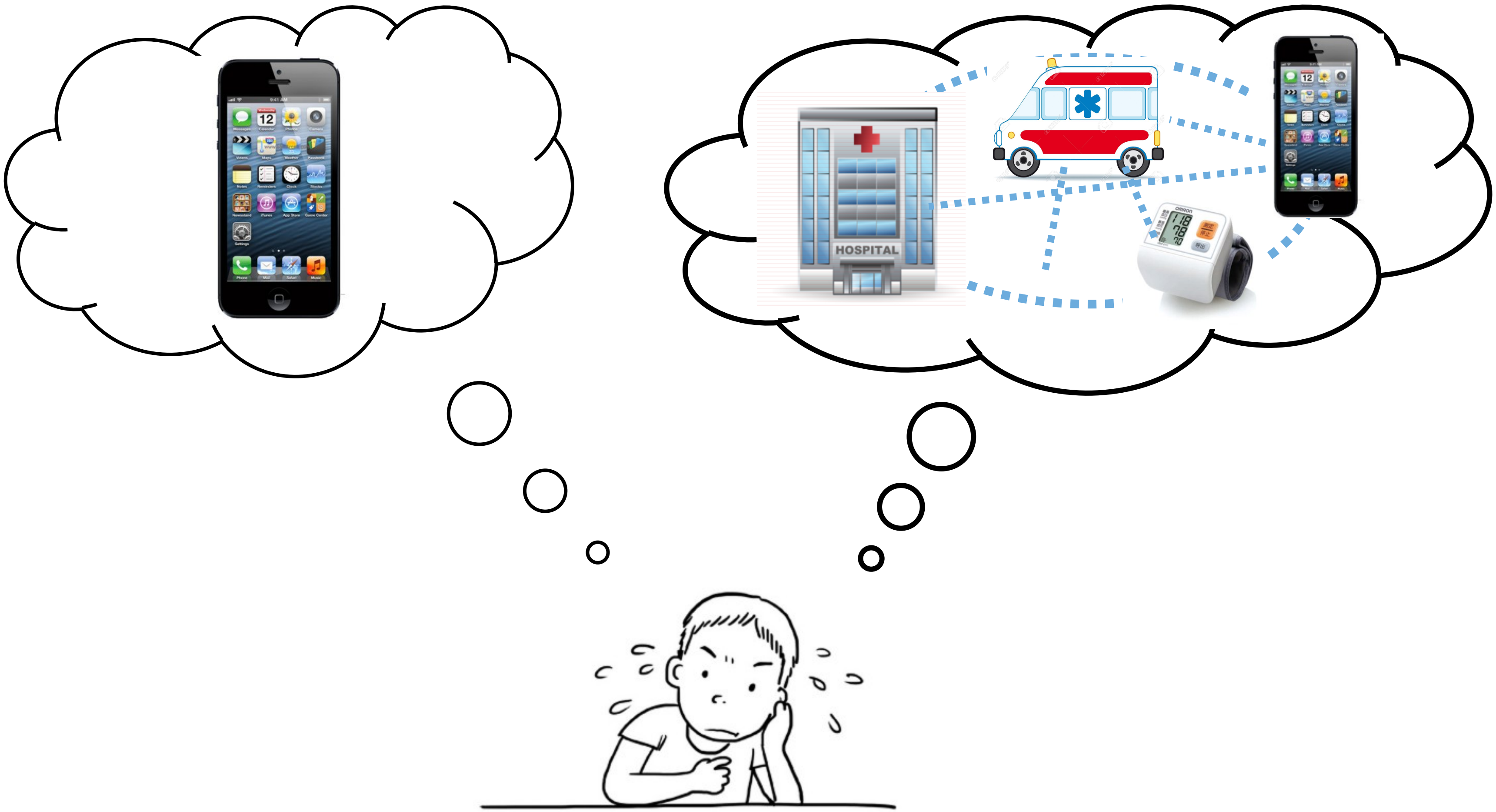
Soojin Park, SIAT, Sogang University, Seoul, South Korea

(psjdream@sogang.ac.kr)

Young B. Park, Department of Computer Science & Engineering, Dankook University, Seoul, South Korea (ybpark@dankook.ac.kr)

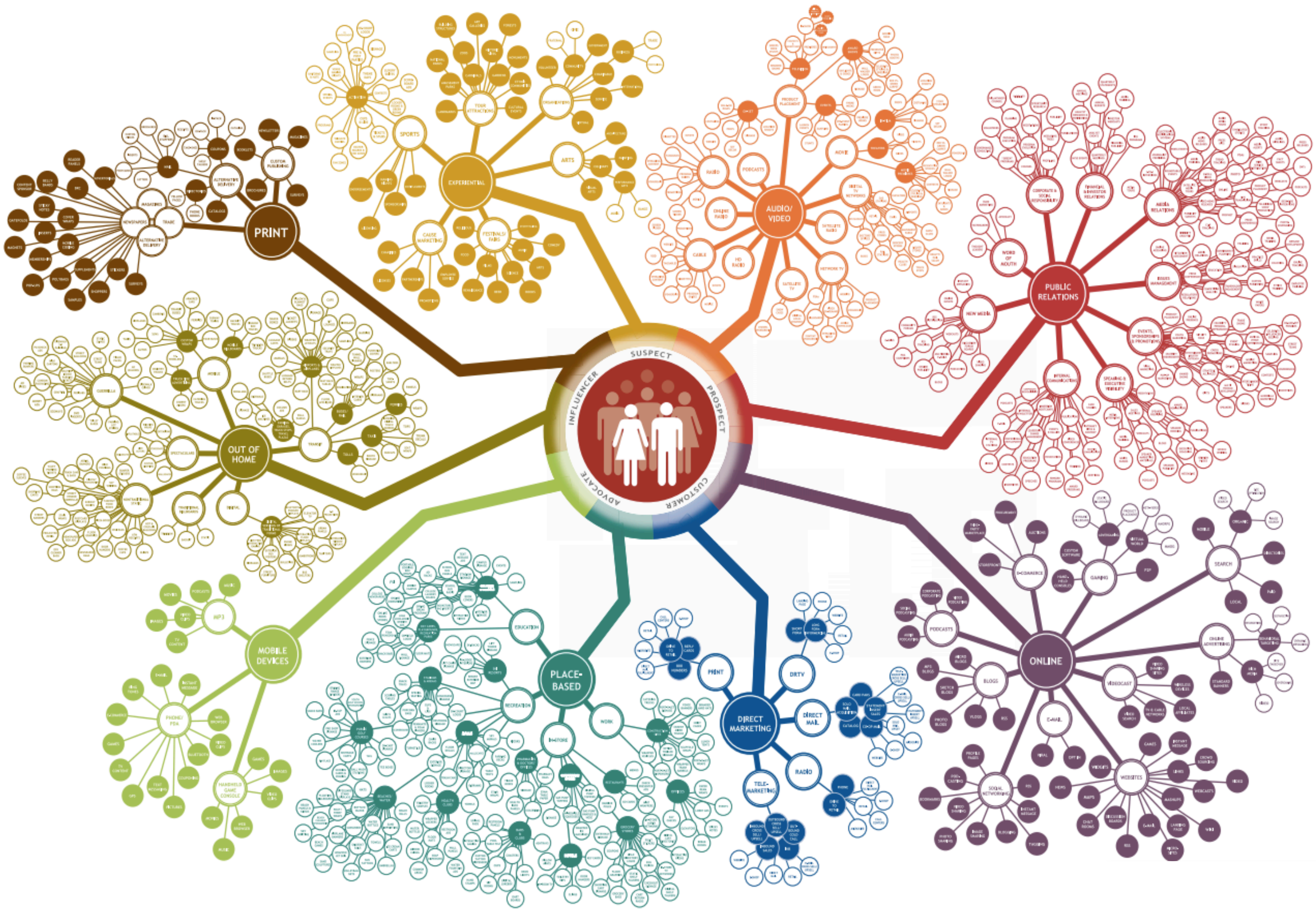
Research Background

Paradigm : A Single System → System of Systems

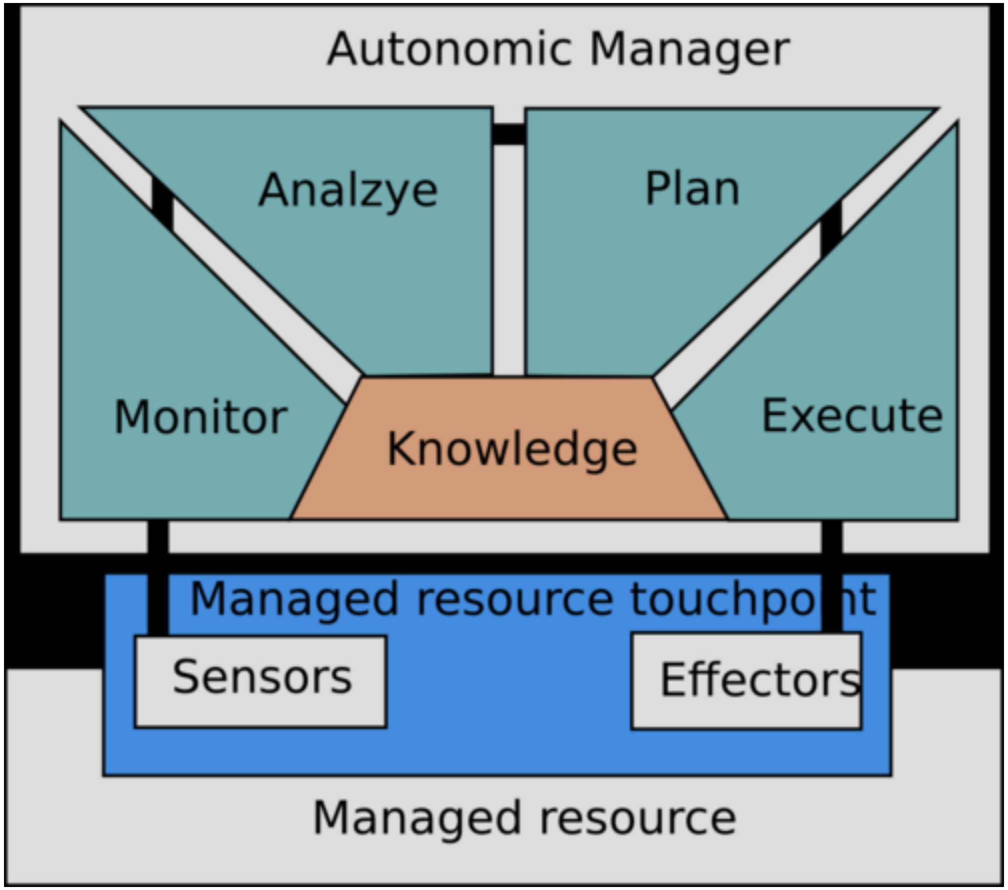


IT Ecosystem

An **IT Ecosystem** is a complex system compound composed of interactive and autonomous individual systems, adaptive as a whole based on local a daptivity [1].



MAPE-K loop


Autonomic Manager
Analzye
Plan
Monitor
Knowledge
Execute
Managed resource touchpoint
Sensors
Effectors
Managed resource

Related Work

Related Work

- Rainbow [5]
  - Introduces a reusable infrastructure as to separate concerns between adaptati on and application logic
  - Provides architecture-based self-adaptability
  - Limited to use when situation-specific action rules are applicable.
- MUSIC [6]
  - Combines previous component-based development methods with Service Ori ented Architecture (SOA)
  - Responds to the distributed and dynamic requirements in mobile environment s
  - Limited in its need for manual adaptation plan update or replacement because the framework does not include goal management features in its MAPE-K self -adaptation layers
- DiVA [7]
  - Provides methodologies and framework for developing self-adaptive systems and for managing variability of self-adaptive systems
  - Based on the characteristics of aspect-oriented programming

Differentiation Points

- Common Limitation of Related Work
  - Self-adaptation is limited to single systems with focus on local adapta tion
  - They are dealing with a MAPE-K loop in a single system

- We propose…
  - A reference architecture framework to support **not a single system but an IT ecosystem.**
  - **The proposed reference architecture framework** includes:
    - **An orchestration mechanism** to select an optimal configuration agains t environment contexts
    - The selection is based on **a cost-benefit model** to evaluate each config uration of participatory systems
    - To minimize the overhead of running the MAPE-K loop, we adopt **geneti c algorithms** in generating candidate configurations

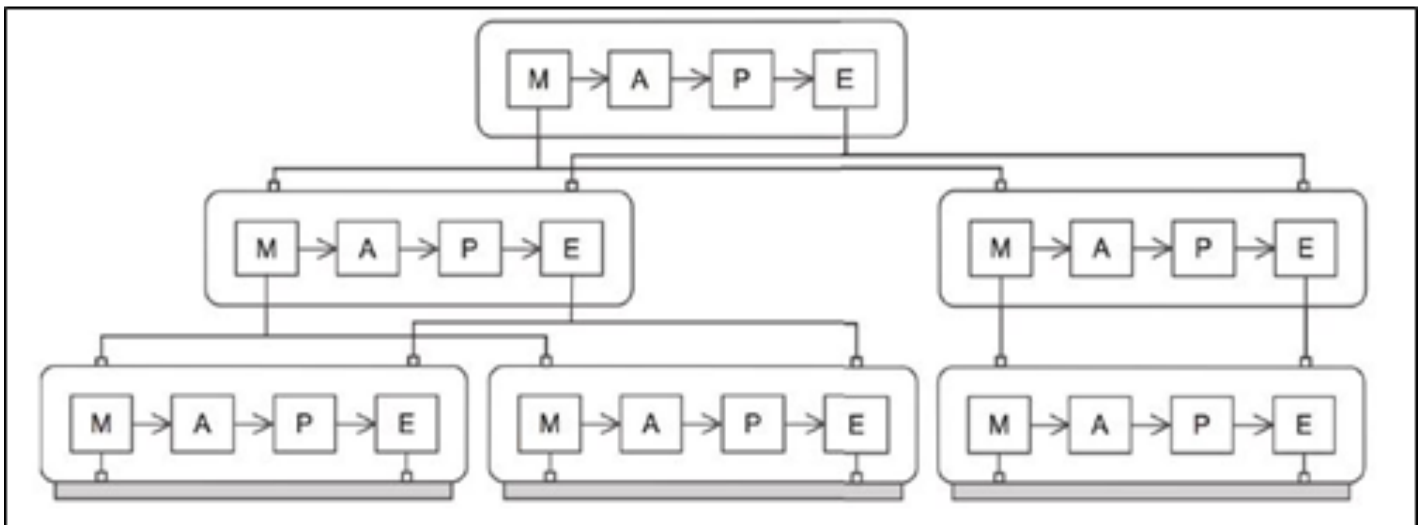
An Overview of *ITE Arbitrator*

Requirements

- Two different kinds of goal models should be provided to specify
  - The **autonomy aspect** of **individual participant systems** in the IT ecosystem
  - The **controllability aspect** of **the entire IT ecosystem**
- Two different loops for achieving the two different types of goals sh ould be supported
  - **N number of MAPE-K loops** to control individual participants syste ms
  - **A MAPE-K loop** to control the entire IT ecosystem
- The role of executing the different loops must be explicitly separate d

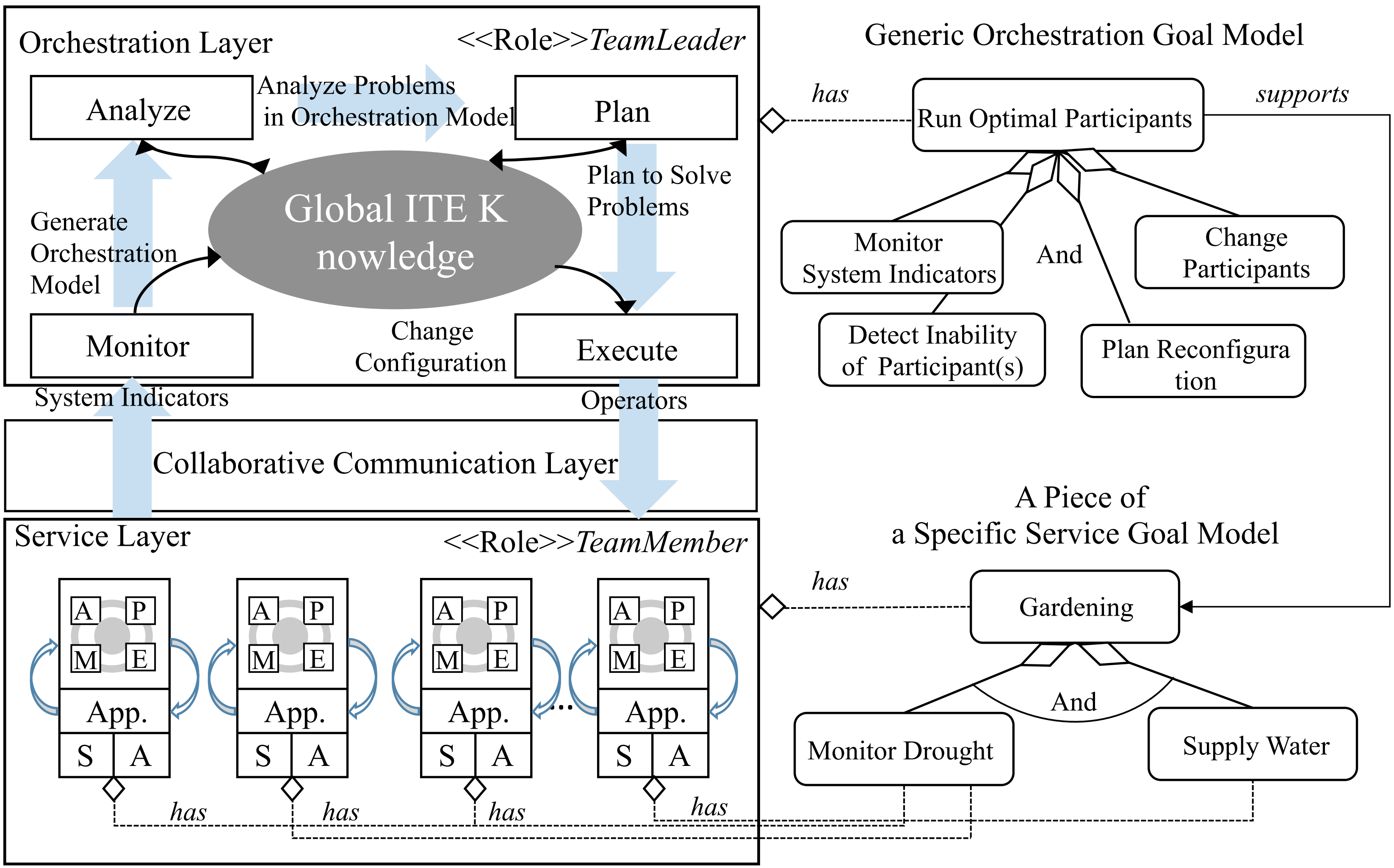
Decentralization Pattern for Multiple MAPE-K Loops

- Hierarchical Control Pattern [3]
  - Motivation
    - The loops can work at different time scales and manage different kind of resour ces, and resources with different localities
    - Control loops need to interact and coordinate actions to avoid conflicts and pro vide certain guarantees about adaptations
  - Solution : provides a layered separation of concerns to manage the complexity of self-adaptation

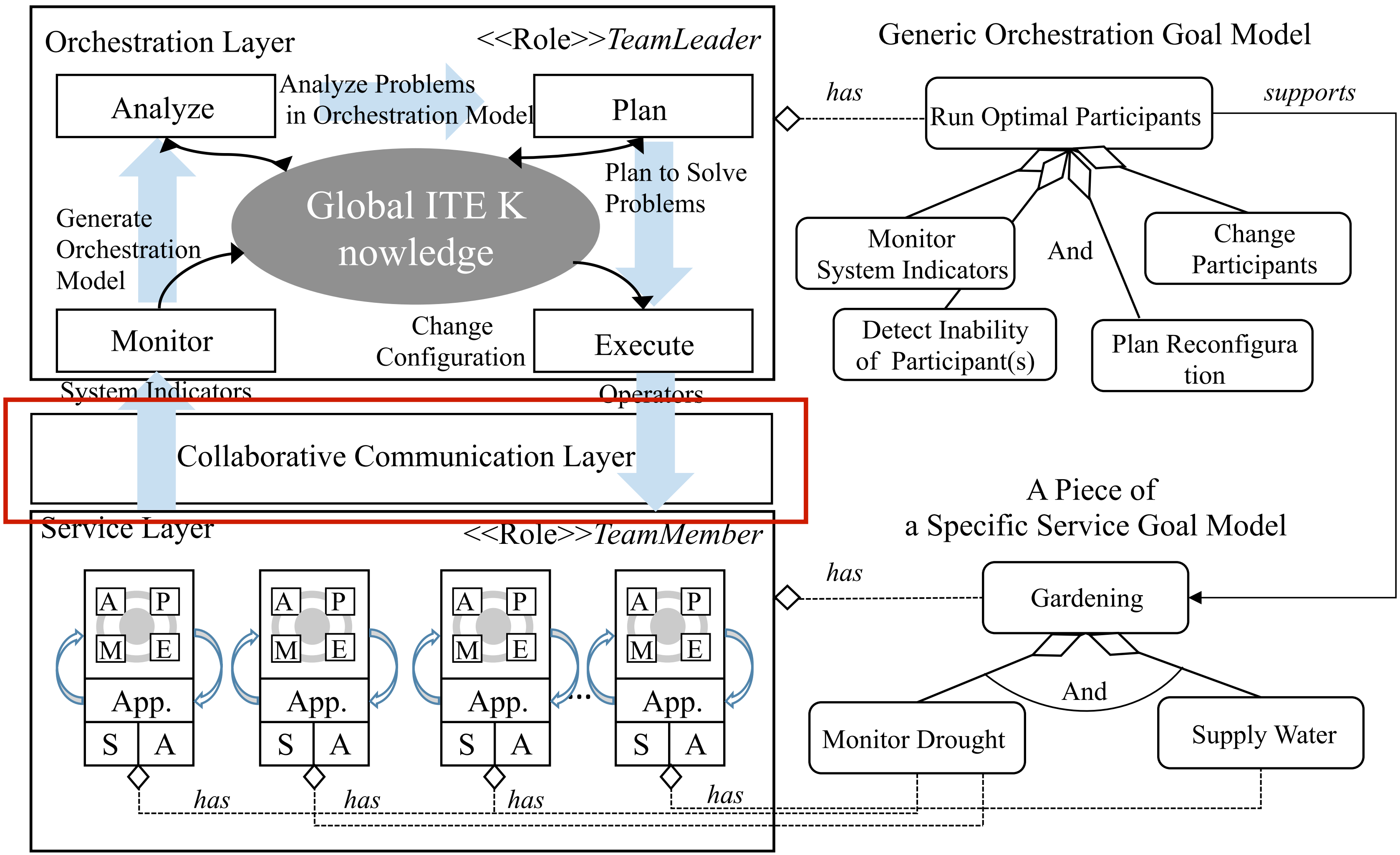


Weyns, D., Schmerl, B., Grassi, V., Malek, S., Mirandola, R., Prehofer, C., Wuttke, J., et al., 2012. On patterns for decentralized control i n self-adaptive systems. In *Software Engineering for Self-Adaptive Systems II. LNCS, Vol. 7475,* 76-107. Springer, Heidelberg. DOI =http: //doi.acm.org/10.10 07/978-3-642-35813-5_4.

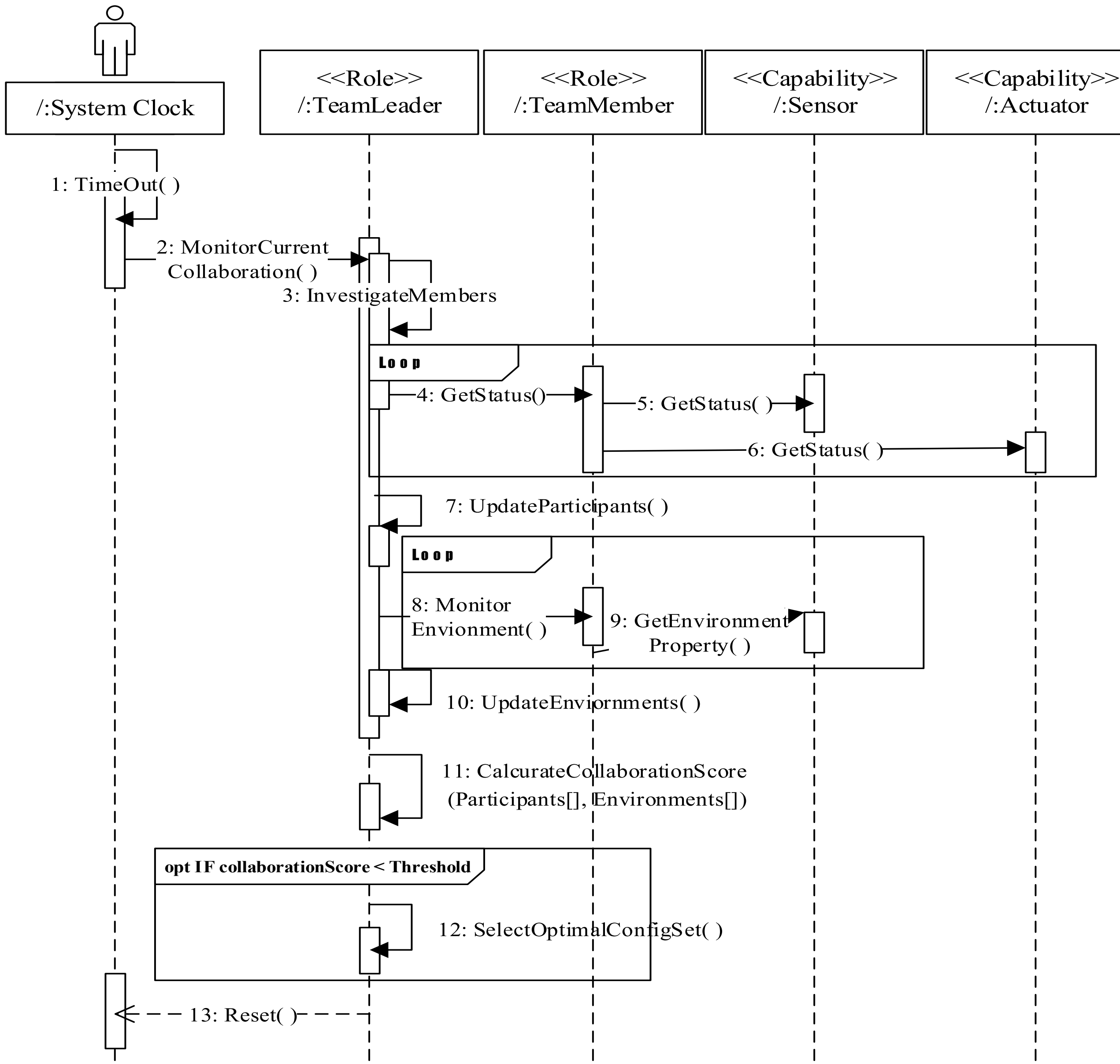
Conceptual Architecture of *ITE Arbitrator*



Conceptual Reference Architecture of *ITE Arbitrator*

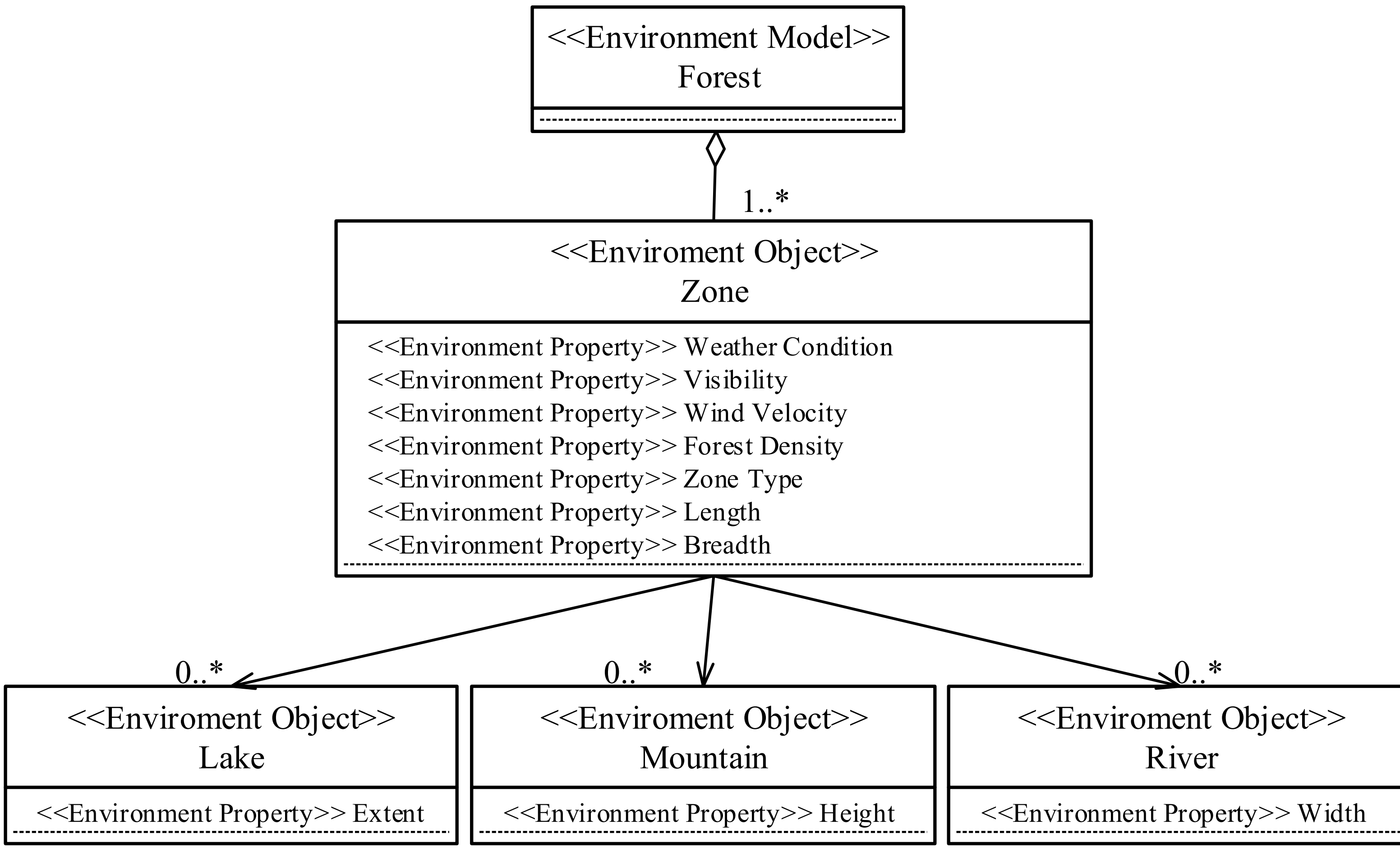


Conceptual Reference Architecture of *ITE Arbitrator*



A Case Study Scenario: The Unmanned Forest Management IT Ecosystem(UFM IT Ecosystem)

UFM IT Ecosystem: Environment Model



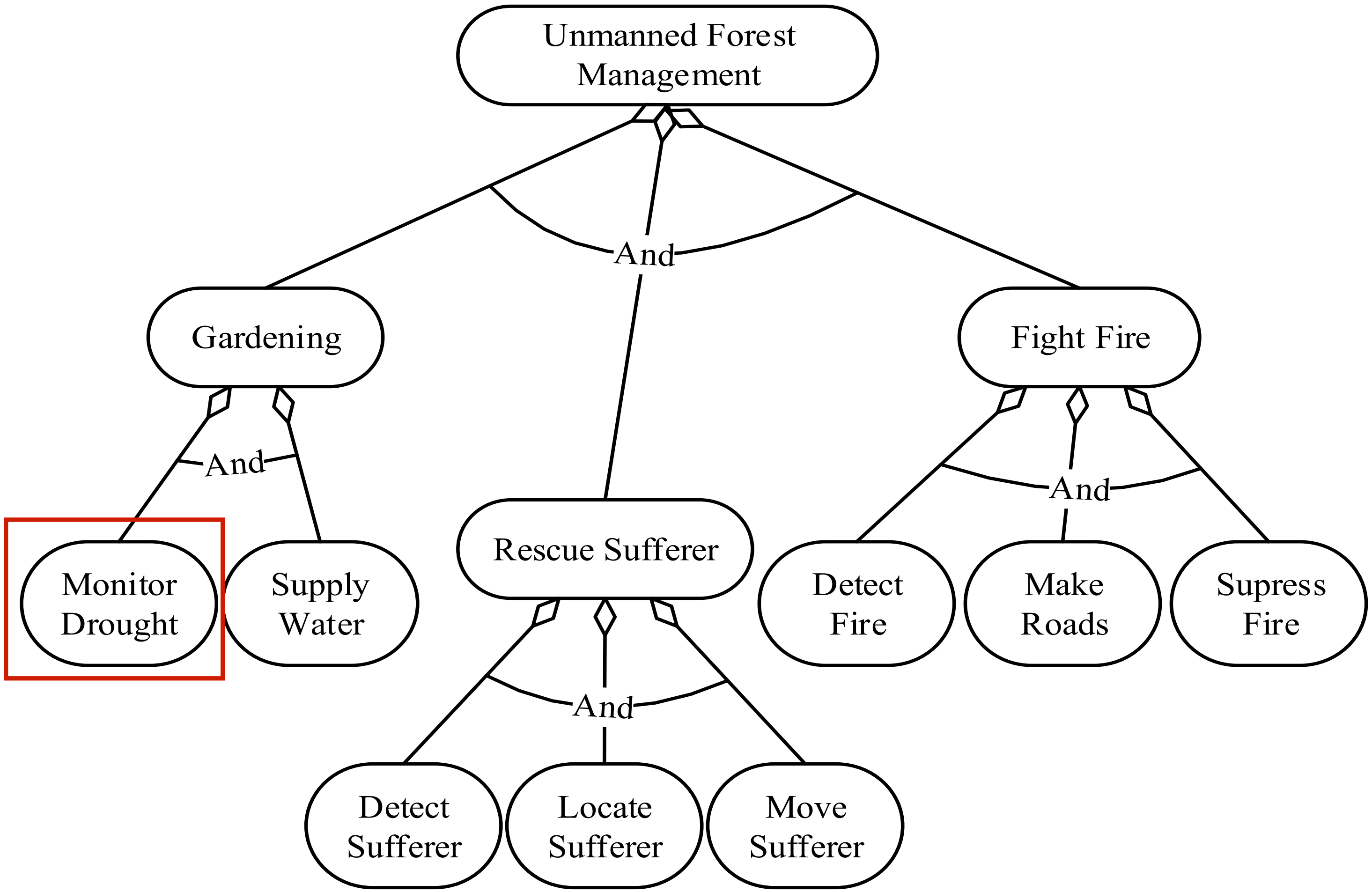
Environment Model: Forest

UFM IT Ecosystem: Environment Model

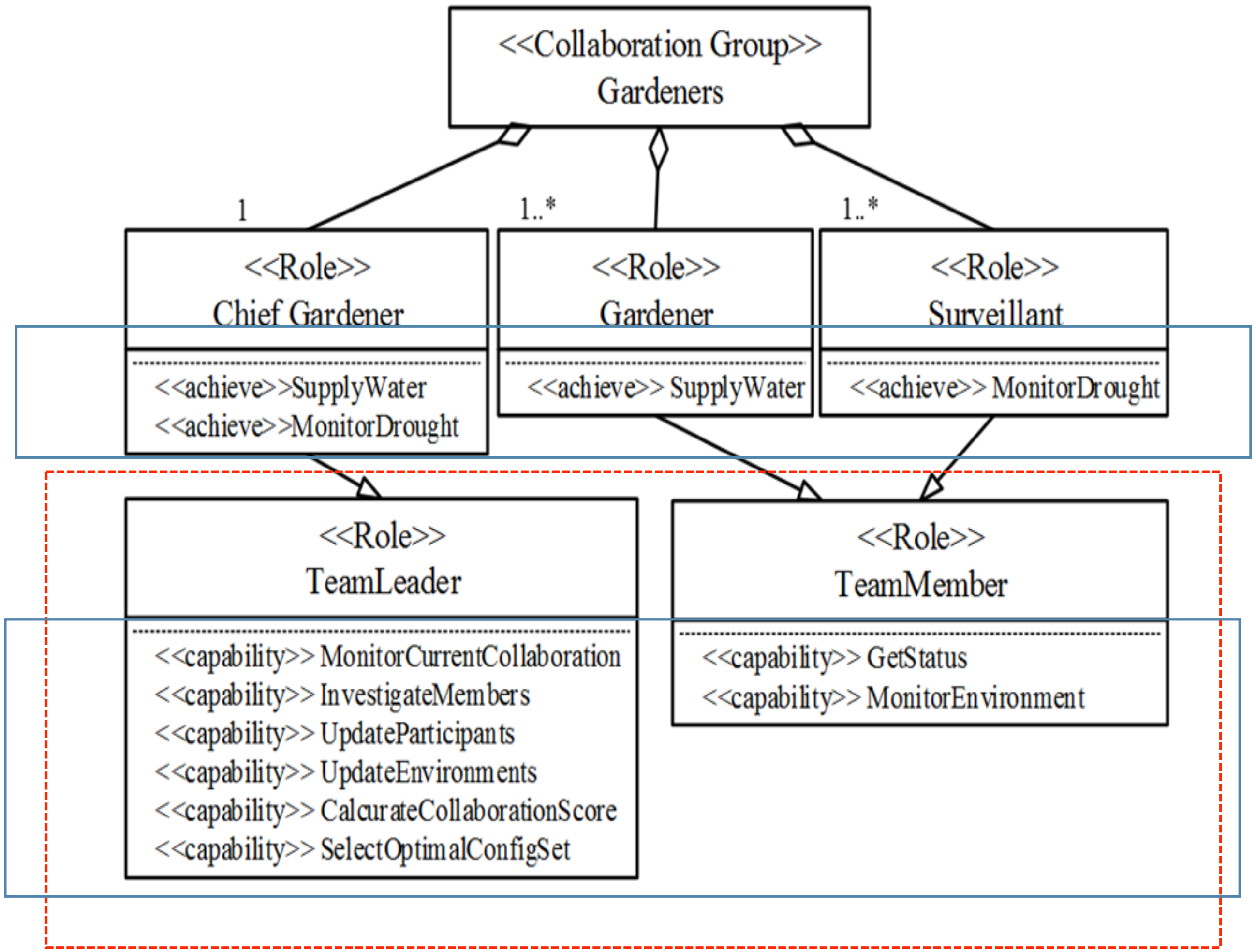
	[0]	[1]	[2]	
Zone Type	has Field	has Field	has Mountain	[0]
Weather Condition	Cloudy	Cloudy	Rainy	
Visibility	10 Km	8 Km	7 Km	
Wind Velocity	5 m/s	8 m/s	15 m/s	
Forest Density	Low	High	Low	
Zone Type	has Field	has Lake	has Mountain	[1]
Weather Condition	Cloudy	Cloudy	Rainy	
Visibility	11 Km	4 Km	3 Km	
Wind Velocity	5 m/s	25 m/s	21 m/s	
Forest Density	Low	Medium	Medium	
Zone Type	has River	has River	has Field	[2]
Weather Condition	Sunny	Sunny	Cloudy	
Visibility	23 Km	16 Km	11 Km	
Wind Velocity	5 m/s	22 m/s	15 m/s	
Forest Density	High	Medium	High	

Instances of a Forest

UFM IT Ecosystem: Service Goal Model



UFM IT Ecosystem: Instantiated Roles

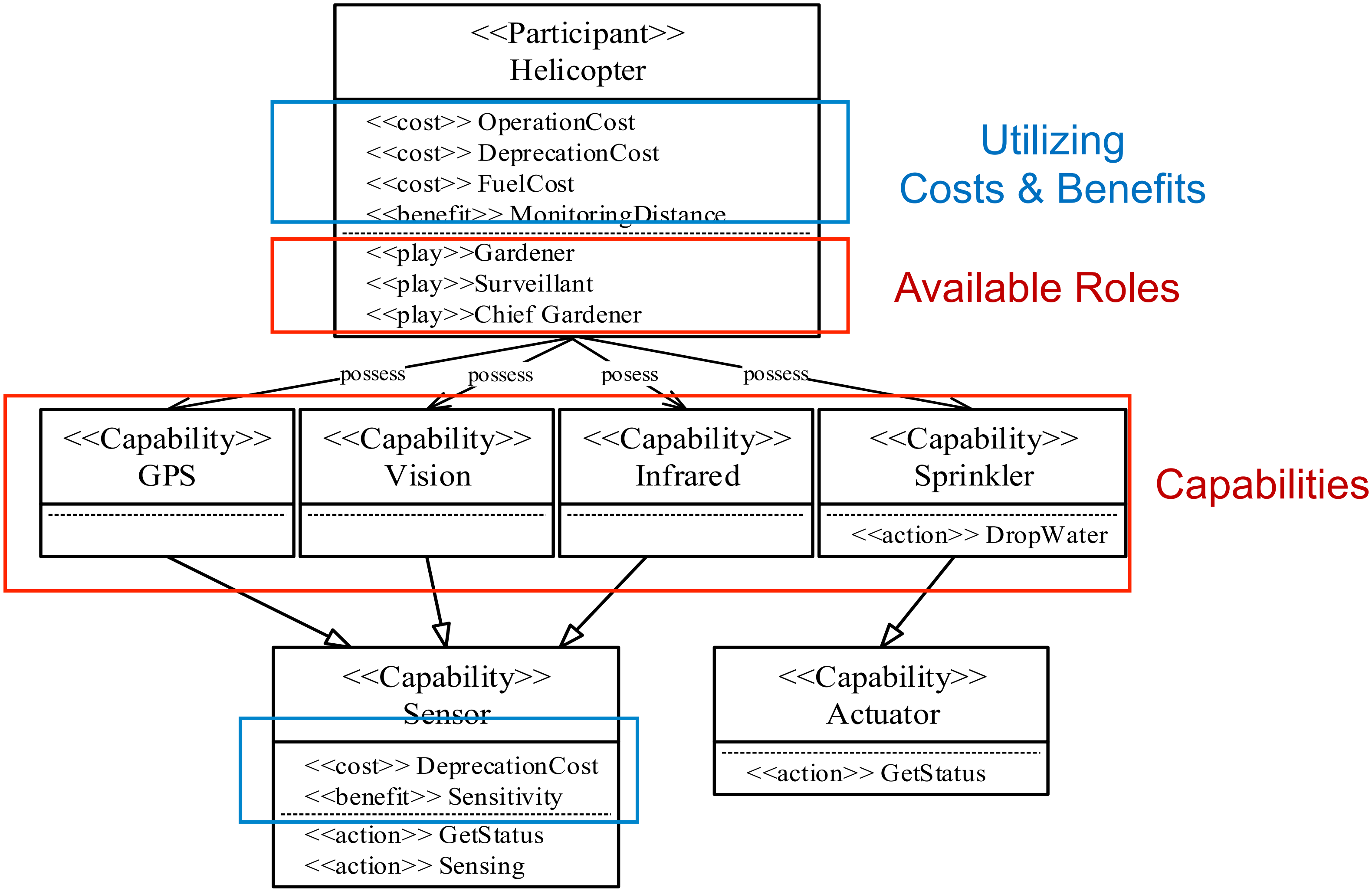


Service goals for UFM IT Ecosystem

Generic capabilities for orchestration goals

Generic Roles In ITE Arbitrator

UFM IT Ecosystem: An Example of Participant System



UFM IT Ecosystem: An Example of Profiles of Participating Unmanned Vehicles and UAVs

		Cost						Benefit							O	P
ID	Type	A	B	D	E	F	G	H	I	J	K	L	M	N		
JE1	Jeep	40	40	14	60	32	19	0.7	0.6	0.64	100	87	74	46	30	80
JE2	Jeep	40	40	15	35	37	22	0.59	0.35	0.73	100	78	54	54	25	69
JE3	Jeep	40	40	12	64	39	23	0.98	0.64	0.78	100	96	57	47	27	63
JE4	Jeep	40	40	13	76	35	21	1	0.76	0.7	100	86	58	46	37	79
JE5	Jeep	40	40	16	45	38	23	0.8	0.45	0.75	100	78	68	38	25	74
JE6	Jeep	40	40	16	85	47	28	0.93	0.85	0.93	100	84	72	51	31	84
UAV1	UAV	0	100	70	86	50	30	0.9	0.86	1	100	100	100	100	102	72
UAV2	UAV	0	100	75	100	50	30	0.95	1	1	100	100	100	100	86	68
HE1	Hlct	80	70	67	75	34	20	0.67	0.75	0.68	100	87	100	74	120	62
HE2	Hlct	80	70	56	60	43	26	0.87	0.6	0.85	100	87	100	78	116	83
AP1	AP	100	80	80	63	37	22	0.62	0.63	0.74	100	87	100	86	170	73
AP2	AP	100	80	83	73	47	28	0.84	0.73	0.93	100	83	100	87	150	86

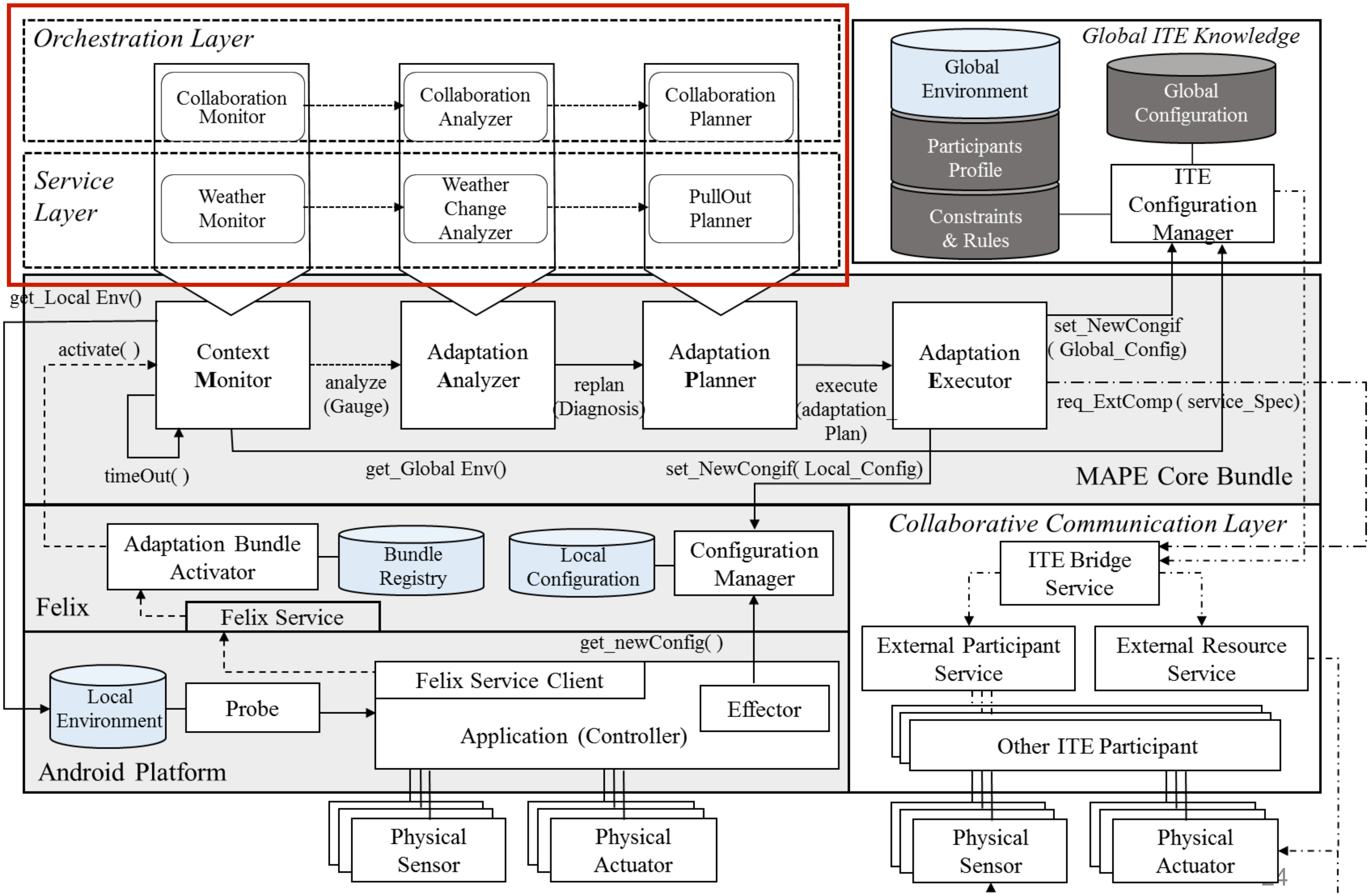
(A) Cost for operating the unmanned vehicle system by human ($/hr)
(B) Deprecation cost ($/hr)
(D) Fuel expenses ($/hr)
(E)~(G) Deprecation cost of Sensors ($/hr): (E) vision sensor, (F) infrared sensor, and (G) GPS sensor.
(H)~(J) Sensitivity of Sensors(0..1): The normalized sensitivity degree of sensors: (H) vision sensor, (I) infrared sensor, and (J) GPS sensor.
(K)~(N) Monitoring Scope (km^2 /hr): It is different according to the geographic property and weather condition shown in ForestZone[ ][ ].
(O) Remained Fuel Quantity (*l*)
(P) Availability Degree (%)

UFM IT Ecosystem: Constraints and Rules

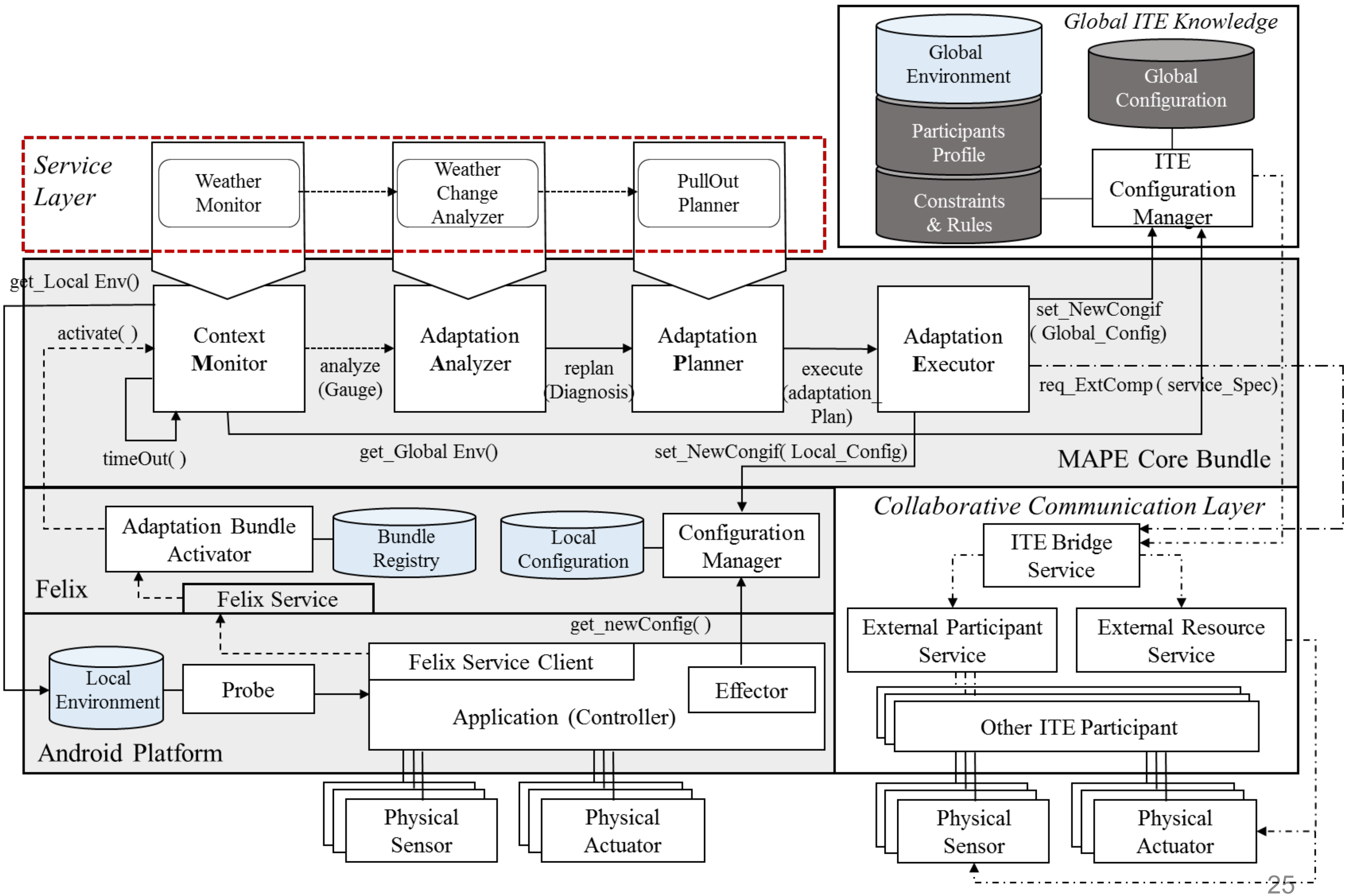
- Violation Rules for Assignment of Participants
  - **IF** (*zone[ ][ ].visibility < 5 (km)*) **THEN** *Helicopter* **CANNOT PLAY AS** *Surveillant* **AT** *zone[ ][ ]*
  - **IF** *(zone[ ][ ].windVelocity > 20 (m/s))* **THEN** *(UAV & Helicopter & AirPlane)* **CANNOT PLAY AS** *Surveillant* **AT** *zone[ ][ ]*
  - **IF** *(zone[ ][ ].weatherCondition = Snowy)* **THEN** *Jeep* **CANNOT PLAY AS** *Surveillant* **AT** *zone[ ][ ]*
  - **IF** *(zone[ ][ ].type = hasLake or hasRiver)* **THEN** *Jeep* **CANNOT PLAY AS** *Surveillant* **AT** *zone[ ][ ]*
  - **IF** *(zone[ ][ ].forestDensity = High) & (zone[ ][ ]. Participant)= Jeep)* **THEN** *Jeep should be operated by a human driver*
  - **IF** (participant. remainedFuelQuantity < Required Quantity for Running One Period) **THEN** Participant **CANNOT PLAY** any role **AT** any zone

An Instantiated Architecture of *ITE Arbitrator*
: UFM IT Ecosystem

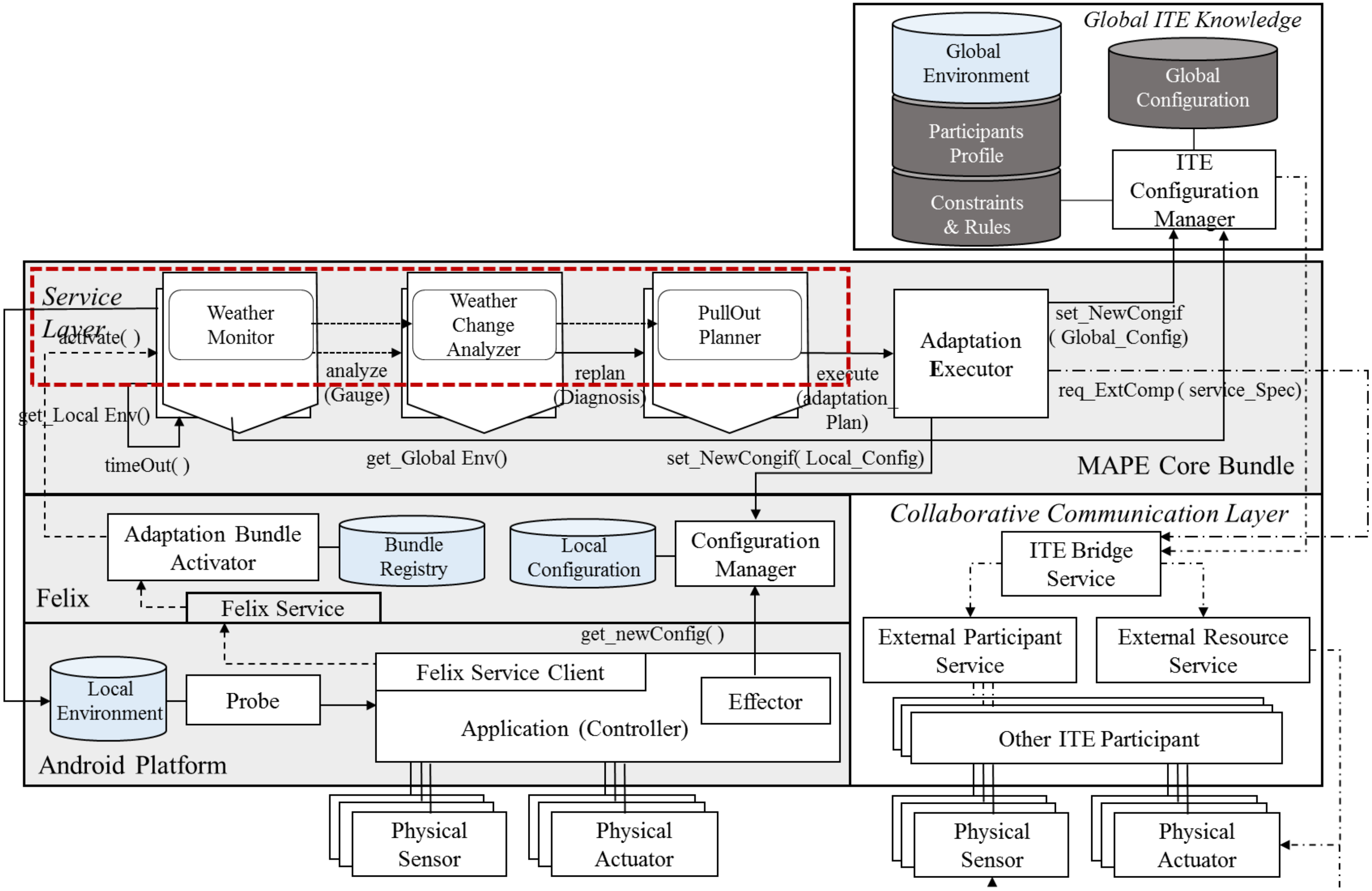
An Instantiated ITE Arbitrator for UFM IT Ecosystem



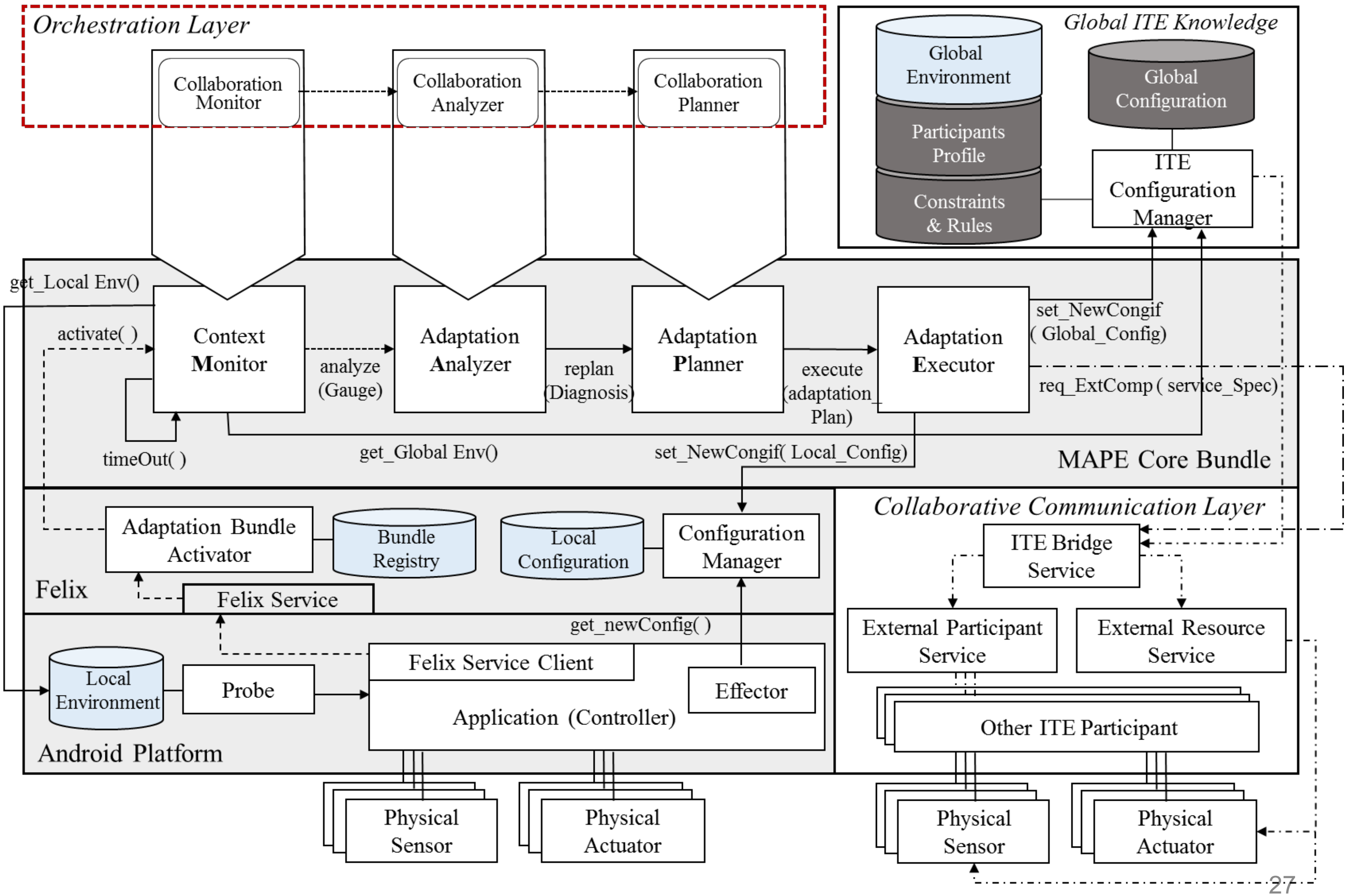
An Instantiated Architecture for *Team Member*



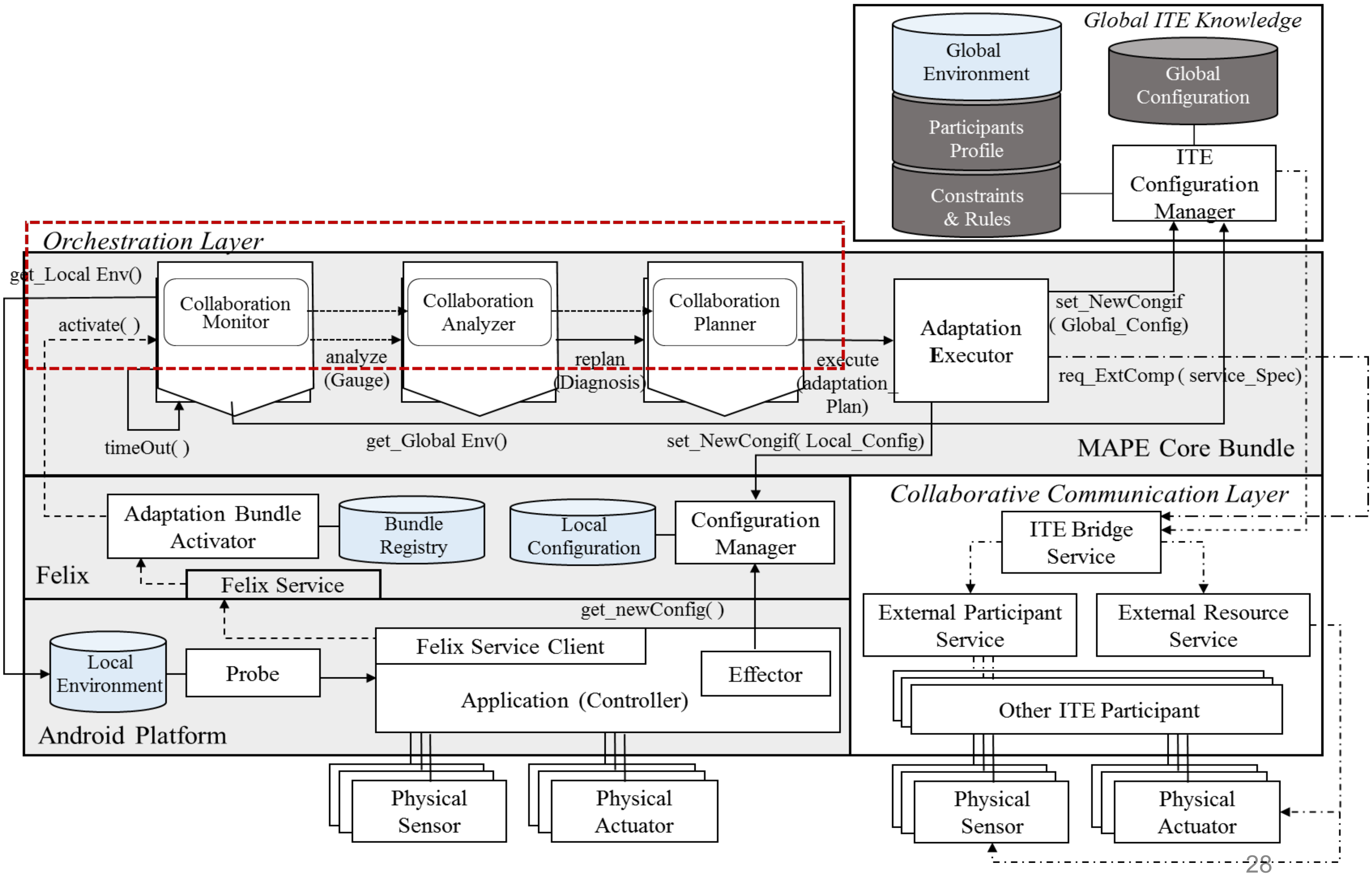
An Instantiated Architecture for *Team Member*



An Instantiated Architecture for *Team Leader*



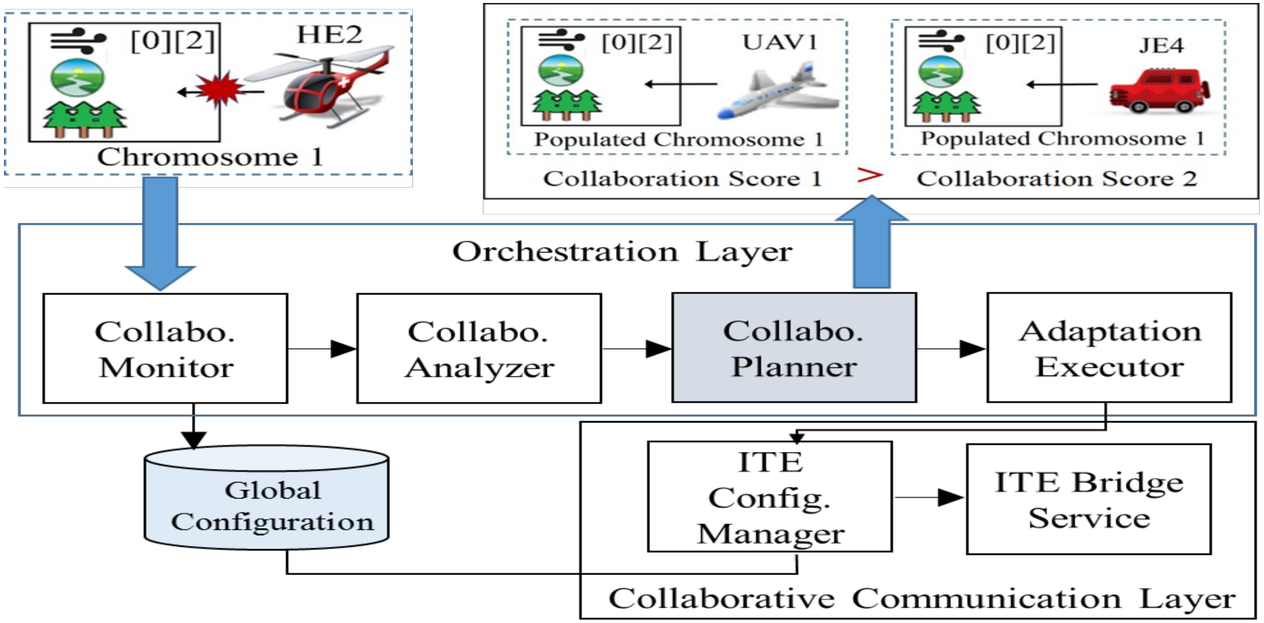
An Instantiated ITE Arbitrator for UFM IT Ecosystem



Orchestration Mechanism of ITE Arbitrator for UFM IT Ecosystem

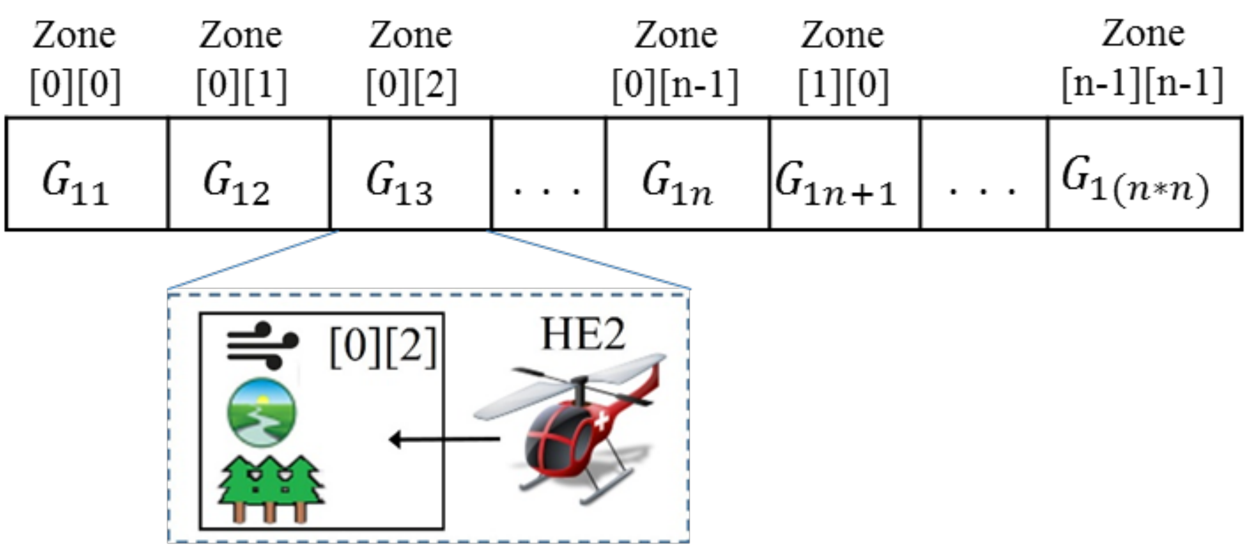
MAPE-K loop for Orchestration Layer of UFM IT Ecosystem

1. (HE2, zone[0][2]) commits violation due to the occurrence of tubulence
2. *Collaboration Monitor*: Generates the global collaboration score from the current global configuration
3. *Collaboration Analyzer*: Compares the current score with the threshold and identifies the invalid pair(s) to commit violation
4. *CollaborationPlanner*: Generates candidate configurations using a genetic algorithm and selects optimal one among candidates
5. *ITE Configuration Manager*: Updates a new optimal configuration as the current one and sends operations to each participant system



Genetic Algorithm for Downsizing the Set of Candidate Configurations

- A *chromosome*: an array of bits as a standard representation of each cand idate solution
- Bit strings for all kinds of attributes for the environments and the participan t systems are included
- The genetic representation makes it possible to facilitate genetic operatio ns

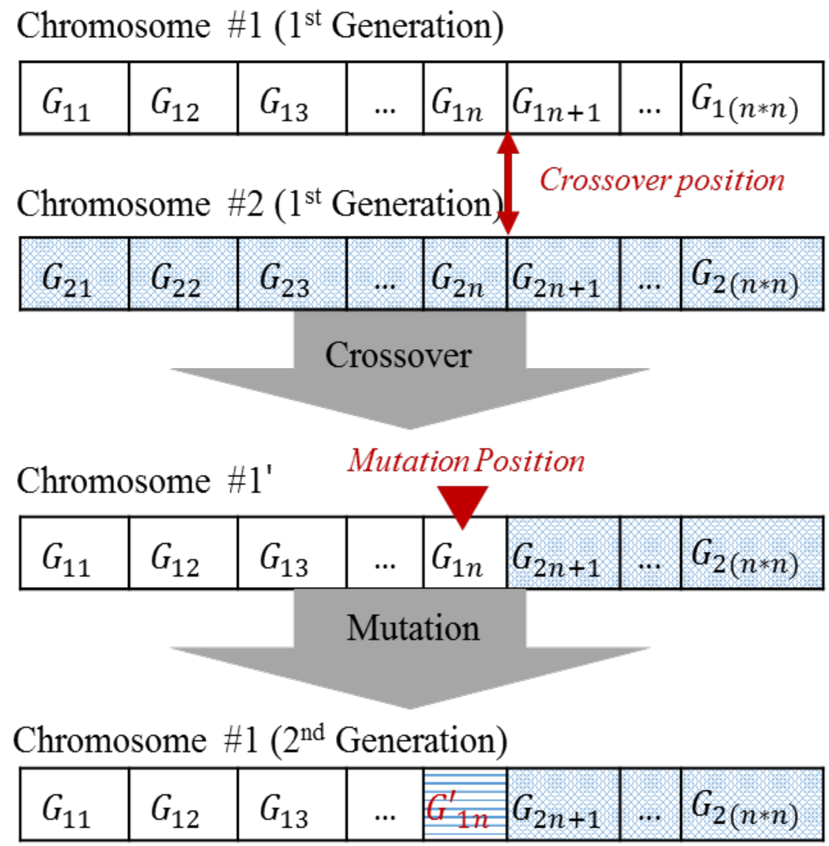


(a) Chromosome Representation for Forest Zone[n][n]

Genetic Algorithm for Downsizing the Set of Candidate Configurations

- If the violation occurs in the curr ent configuration, the genetic al gorithm uses mutation and a cr ossover mechanism to generat e another population.

- Since this next population is ge nerated from the base populatio ns, the size of the next generati on can be controlled and the se arching space can be limited to find the most optimal solution



(b) Propagation of the Next Generation of Chromosome #1

Evaluation of Benefits from Using the Genetic Algorithm

- Experimental Design
  - Target domain: UFM IT Ecosystem
  - 3 x 3 forest zones(9 zones)
  - 12 candidate participant systems
  - 30 times of reconfiguration has been occurred

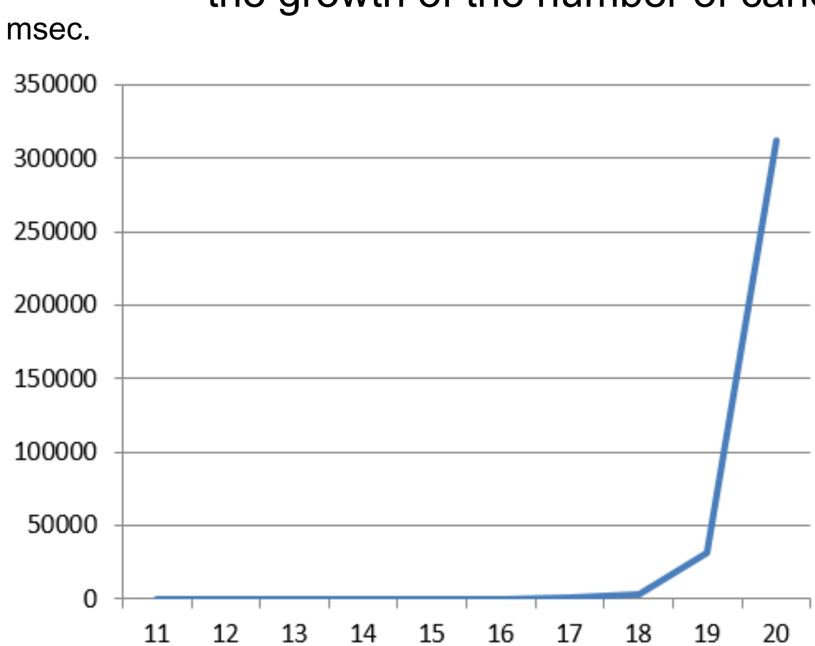
- Without the Genetic Algorithm
  - Average elapsed time: 43 sec.
  - Increasing the number of candidate participant systems:

    - 13 candidates → 39 minutes

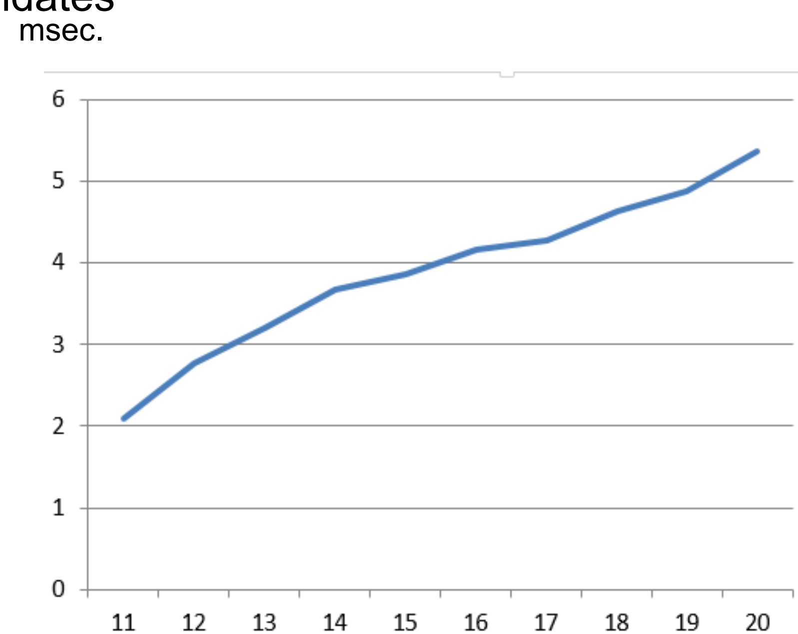
    - 14 candidates → it doesn't work

Evaluation of Benefits from Using the Genetic Algorithm

- Using the Genetic Algorithm
  - Average elapsed time: 20 msec. (43 sec. → 20 msec.)
  - Increasing the number of candidate participant systems
    - Average elapsed time for selecting the first configuration is exponentially increased when the candidates are over 20
    - Average elapsed time for each re-configuration is gradually ascended against the growth of the number of candidates



Average elapsed time for
the first configuration



Average elapsed time for
each re-configuration

Global Collaboration Score

- ***Collaboration score*** as a fitness function
  - A higher collaboration score for a chromosome indicates that chromosome's greater effectiveness in the mapping between the paired participant and the environment.

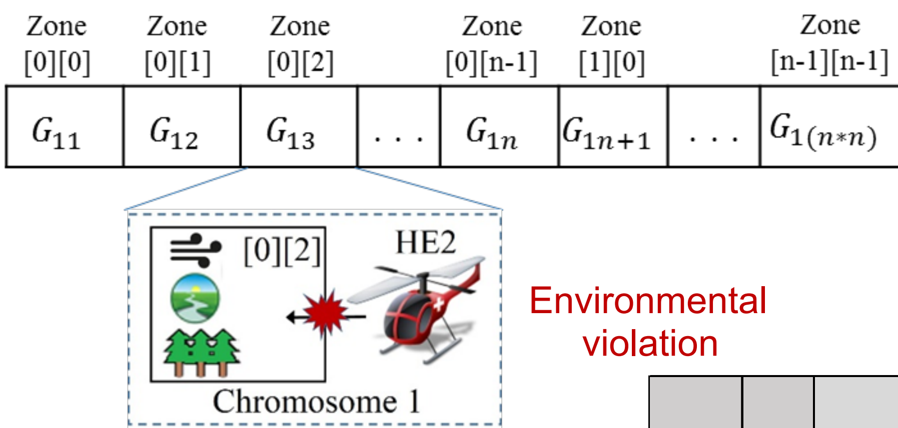
- Utility Functions

$$\left.\begin{aligned}
&cost(p_i, e_j) = \sum_k(\eta_k \times Ncf_k) \quad \text{where } \sum_k \eta_k = 1\\
&benefit(p_i,\ e_j) = Nbf_k = \frac{(Bf_k - Bf_{min})}{(Bf_{max} - Bf_{min})}\\
&\text{score}(Conf_r) = w_0 \times \text{benefit}(Conf_r) + w_1 \times cost(Conf_r)\\
&\qquad where \sum |\omega_i| = 1 \ and \prod \omega_i = -1
\end{aligned}\right\} \text{Local optimization}$$

$$Global\ Collaboration\ Score = \sum_j C.Score(p_i, e_j)$$

$$\text{where } (p_i, e_j) \text{ } is\ optimal\ chromosome\ for\ e_j$$

Global Collaboration Score



Environmental violation

$$cost(p_i, e_j) = \sum_k (\eta_k \times Ncf_k) \quad \text{where } \sum_k \eta_k = 1$$

$$benefit(p_i, e_j) = Nbf_k = \frac{(Bf_k - Bf_{min})}{(Bf_{max} - Bf_{min})}$$

$$score(Conf_r) = w_0 \times benefit(Conf_r) + w_1 \times cost(Conf_r)$$

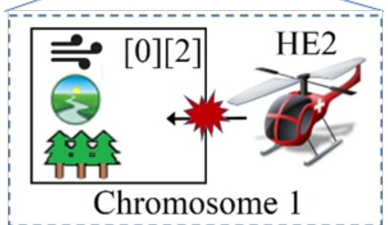
$$where \sum |\omega_i| = 1 \; and \prod \omega_i = -1$$

	[0]	[1] p_i		
Zone Type	has Field	has Field	ha	
Weather Condition	Cloudy	Cloudy		
Visibility	10 Km	8 Km		
Wind Velocity	5 m/s	8 m/s		
Forest Density	Low	High		
Zone Type	has Field	has Lake	ha	
Weather Condition	Cloudy	Cloudy		
Visibility	11 Km	4 Km		
Wind Velocity	5 m/s	25 m/s		
Forest Density	Low	Medium	Medium	
Zone Type	has River	has River	has Field	
Weather Condition	Sunny e_j	Sunny	Cloudy	
Visibility	23 Km	16 Km	11 Km	[2]
Wind Velocity	5 m/s	22 m/s	15 m/s	
Forest Density	High	Medium	High	

		Cost						Benefit							O	P
ID	Type	A	B	D	E	F	G	H	I	J	K	L	M	N		
JE1	Jeep	40	40	14	60	32	19	0.7	0.6	0.64	100	87	74	46	30	80
JE2	Jeep	40	40	15	35	37	22	0.59	0.35	0.73	100	78	54	54	25	69
JE3	Jeep	40	40	12	64	39	23	0.98	0.64	0.78	100	96	57	47	27	63
JE4	Jeep	40	40	13	76	35	21	1	0.76	0.7	100	86	58	46	37	79
JE5	Jeep	40	40	16	45	38	23	0.8	0.45	0.75	100	78	68	38	25	74
JE6	Jeep	40	40	16	85	47	28	0.93	0.85	0.93	100	84	72	51	31	84
UAV1	UAV	0	100	70	86	50	30	0.9	0.86	1	100	100	100	100	102	72
UAV2	UAV	0	100	75	100	50	30	0.95	1	1	100	100	100	100	86	68
HE1	Hlct	80	70	67	75	34	20	0.67	0.75	0.68	100	87	100	74	120	62
HE2	Hlct	80	70	56	60	43	26	0.87	0.6	0.85	100	87	100	78	116	83
AP1	AP	100	80	80	63	37	22	0.62	0.63	0.74	100	87	100	86	170	73
AP2	AP	100	80	83	73	47	28	0.84	0.73	0.93	100	83	100	87	150	86

Global Collaboration Score

Zone [0][0]	Zone [0][1]	Zone [0][2]		Zone [0][n-1]	Zone [1][0]		Zone [n-1][n-1]
G_{11}	G_{12}	G_{13}	. . .	G_{1n}	G_{1n+1}	. . .	$G_{1(n*n)}$



Environmental violation

$$cost(p_i, e_j) = \sum_k (\eta_k \times Ncf_k) \quad \text{where } \sum_k \eta_k = 1$$

$$benefit(p_i, e_j) = Nbf_k = \frac{(Bf_k - Bf_{min})}{(Bf_{max} - Bf_{min})}$$

$$score(conf_j) = w_0 \times benefit(conf_r) + w_1 \times cost(conf_r)$$

$$where \sum |\omega_i| = 1 \; and \prod \omega_i = -1$$

$$Global\ Collaboration\ Score = \sum_j C.Score(p_i, e_j)$$

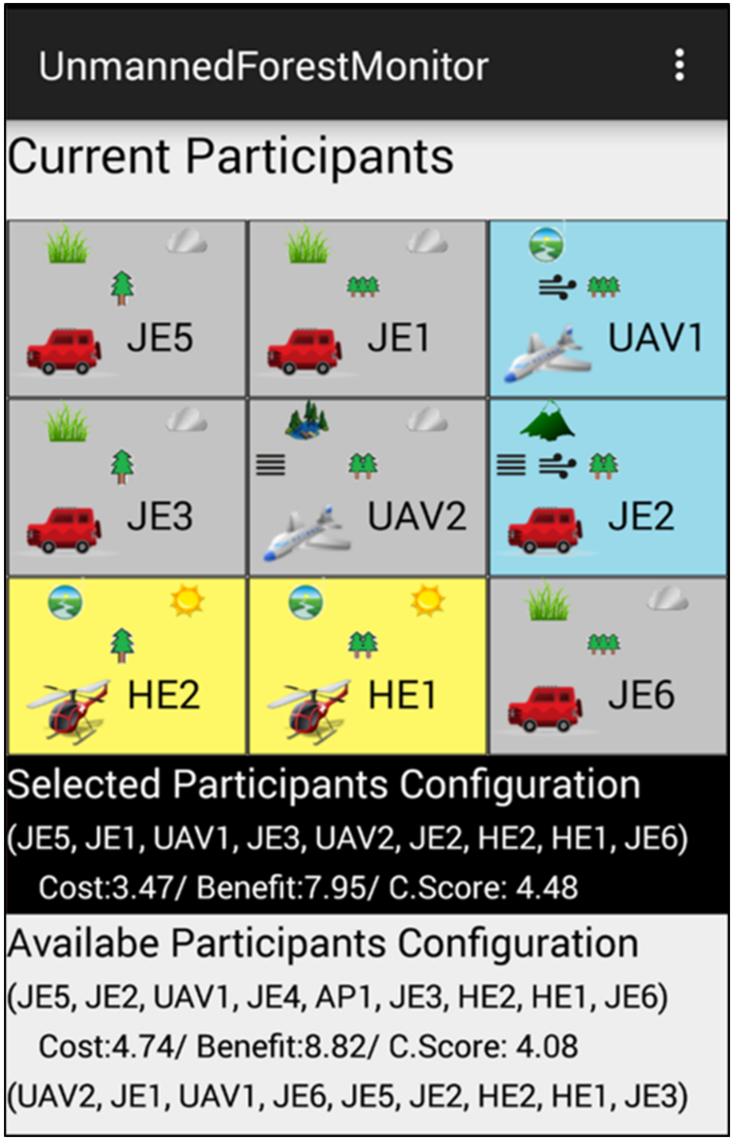
where (p_i, e_j) is optimal chromosome for e_j

	[0] p_i	[1]	
Zone Type	has Field	has Field	
Weather Condition	Cloudy e_j	Cloudy	
Visibility	10 Km	8 Km	
Wind Velocity	5 m/s	8 m/s	
Forest Density	Low	High	
Zone Type	has Field	has Lake	
Weather Condition	Cloudy	Cloudy	
Visibility	11 Km	4 Km	
Wind Velocity	5 m/s	25 m/s	
Forest Density	Low	Medium	
Zone Type	has River	has River	has Field
Weather Condition	Sunny	Sunny	Cloudy
Visibility	23 Km	16 Km	11 Km
Wind Velocity	5 m/s	22 m/s	15 m/s
Forest Density	High	Medium	High

[2]

		Cost						Benefit							O	P
ID	Type	A	B	D	E	F	G	H	I	J	K	L	M	N		
JE1	Jeep	40	40	14	60	32	19	0.7	0.6	0.64	100	87	74	46	30	80
JE2	Jeep	40	40	15	35	37	22	0.59	0.35	0.73	100	78	54	54	25	69
JE3	Jeep	40	40	12	64	39	23	0.98	0.64	0.78	100	96	57	47	27	63
JE4	Jeep	40	40	13	76	35	21	1	0.76	0.7	100	86	58	46	37	79
JE5	Jeep	40	40	16	45	38	23	0.8	0.45	0.75	100	78	68	38	25	74
JE6	Jeep	40	40	16	85	47	28	0.93	0.85	0.93	100	84	72	51	31	84
UAV1	UAV	0	100	70	86	50	30	0.9	0.86	1	100	100	100	100	102	72
UAV2	UAV	0	100	75	100	50	30	0.95	1	1	100	100	100	100	86	68
HE1	Hlct	80	70	67	75	34	20	0.67	0.75	0.68	100	87	100	74	120	62
HE2	Hlct	80	70	56	60	43	26	0.87	0.6	0.85	100	87	100	78	116	83
AP1	AP	100	80	80	63	37	22	0.62	0.63	0.74	100	87	100	86	170	73
AP2	AP	100	80	83	73	47	28	0.84	0.73	0.93	100	83	100	87	150	86

Global Collaboration Score



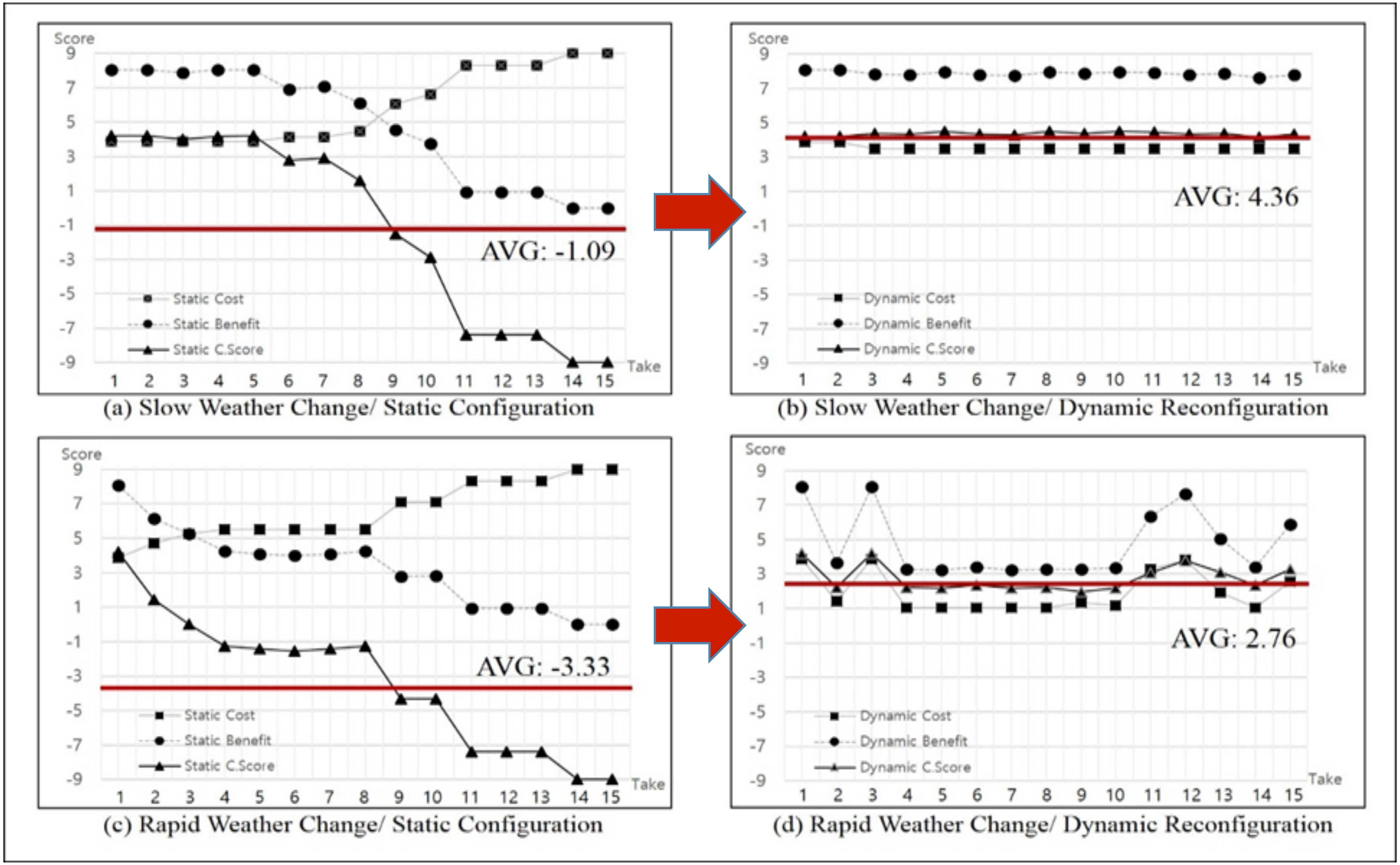
Conclusions & On Going Work

- We have proposed
  - A reference architecture, *ITE Arbitrator*, to support local self-adaptatio n of individual participant systems as well as orchestration mechanis ms across an entire IT ecosystem.
  - A cost-benefit decision model in determining the optimal configuration in response to environmental changes; it also makes efforts to keep computational overhead at an acceptable level, even for scaling syste m configurations through applying genetic algorithms.
- We are doing…
  - Quantitative evaluations to show that the proposed *ITE Arbitrator* help s an IT ecosystem provide sustainable services.
- We plan…
  - To continue to self-evaluate as extend our framework to other domain s of IT Ecosystems.


THANK YOU

Sustainability Enhancement

- Changes in the Global Collaboration Score



CASE: Unmanned Forest Management IT Ecosystem